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Vehicle Summon System, Method And Vehicle Incorporating Same

Abstract

A system for summoning a vehicle includes a portable electronic device, a plurality of communication devices or nodes for installation at the vehicle, and a controller. The controller is configured for establishing wireless communication with the portable electronic device via the plurality of communication devices or nodes. The controller is configured for verifying the portable electronic device as an authorized device associated with the vehicle. The controller is configured for determining a relative location of the portable electronic device via triangulation/trilateration of distances between the portable electronic device and the plurality of communication devices or nodes installed in the vehicle. The controller is configured for calculating a path toward the relative location. The controller is configured for activating a running gear of the vehicle to autonomously execute the path.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to EP 24161536.8 filed Mar. 5, 2024, the entire disclosure of which is incorporated by reference.

FIELD

[0002] The present disclosure relates to a system for summoning a vehicle, by a user that is external to the vehicle.

BACKGROUND

[0003] Many vehicles presently offer enhanced parking features. For example, a limited forward and/or backward movement may be remotely activated to enable an empty vehicle to enter into or exit from a tight parking space where access to a door by the driver is obscured. In a related example, a “vehicle summon” function may enable the vehicle to drive autonomously over a short distance, such as from a parking space toward the waiting position of a user at a roadside, in the manner of a valet. Such functions may be operated from a key fob and/or as part of phone-as-a-key (PaaK) solutions where a smart phone may be enabled with unlocking and activation of the limited self-drive functions.

[0004] Demand for such vehicle features is growing, for the purpose of enhancing user experience, convenience and safety. An example of hailing a vehicle via a mobile system is found disclosed in US2022092718A1.

[0005] The background description provided here is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

SUMMARY

[0006] The present disclosure seeks to provide a vehicle summon system offering improved user experience, e.g. by enabling greater security and/or convenience or, at the least, providing the public and manufacturers with an alternative to known solutions.

[0007] According to a first aspect, a vehicle summon system is provided comprising: a portable electronic device; a plurality of communication devices or nodes, comprising ultra-wide band, UWB, devices, for installation at a vehicle; a controller configured for: establishing wireless communication with the portable electronic device, via the communication devices or nodes; verifying the portable electronic device as an authorized device associated with the vehicle; determining a relative location of the portable device, via triangulation/trilateration of distances between the portable device and the plurality of communication devices or nodes installed in the vehicle; calculating a path toward the relative location; and activating running gear of the vehicle **(10)** to autonomously execute the path.

[0008] In a broader sense the system at least comprises: a portable electronic device, e.g. for carrying by a user; a communication device configured to detect and determine a relative location of the portable electronic device; the system being configured for establishing wireless communication, via the communication device, with the portable electronic device and controlling the interactions thereof to determine the relative location; preferably authenticating the portable electronic device as authorized to enact the system; and calculating a path toward the relative location. Alternatively: a system for summoning a vehicle, the system comprising: a portable

electronic device; a communication device or node for installation at a vehicle; a controller configured for: establishing wireless communication with the portable electronic device, via the communication device or node; verifying the portable electronic device as an authorized device associated with the vehicle; determining a relative location of the portable device, via the communication device or node; calculating a path toward the relative location; and activating running gear of the vehicle to autonomously execute the path.

[0009] A controller may be further configured to mobilize the running gear of a vehicle installed with the system, and thereby guide the vehicle along the calculated path toward the portable electronic device, or at least proximate thereto. In this way, the system is particularly useful for implementation into a vehicle that has partial or fully autonomous driving capability, e.g. operable over short ranges, to “summon” the vehicle to a user.

[0010] The range may be determined by capabilities of the communication device, e.g. an ultra wide band (UWB) device/transceiver. Alternatively, or in addition, range is preferred to be within eyesight of a user for safe operation. In any event, UWB devices are already known to be installed in modern vehicles, e.g. in connection with proximity unlocking and PaaK-type functions. In alternate forms, other communication methods/wireless protocols may be utilized such as Bluetooth™. In principle, any current or future form of ranging/positioning protocol/function may be successfully implemented, or a combination of devices/sensors employed to accurately determine a portable device relative location, by direct communication between the vehicle's onboard devices and the portable device.

[0011] According to the present disclosure, such onboard communication devices/capabilities of the vehicle are given an enhanced role for use in a vehicle summon system.

[0012] In this way, available sensors/equipment in a vehicle are utilized without the need for additional or retrofit components. A control unit can be programmed with the required function to utilize the vehicle's resources and an existing portable device such as a smart phone equipped with compatible features, e.g. UWB and PaaK.

[0013] In embodiments, the controller/system is configured for continual or periodic communication to determine a current relative location of the portable device and to update/recalculate the path toward said current relative location. In this way, if a user carrying the portable device moves, such as to obtain a more advantageous position (e.g. avoiding a puddle or obstacle), then the vehicle is able to be summoned to the updated location.

[0014] A vehicle implementing the system is outlined herein. The vehicle at least comprises: communication devices (e.g. UWB devices) configured for wireless communication with an external portable electronic device; and a processor configured for: establishing and/or controlling wireless communication between the communication devices and the portable electronic device; determining a location of the portable device, relative to the vehicle (e.g. upon receipt of a command issued by the portable device); calculating a path toward the relative location; mobilizing running gear of the vehicle, to thereby guide the vehicle along the path toward the portable electronic device.

[0015] In this way, the vehicle is able to be summoned (e.g. autonomously driven) to a position at or proximate the portable device. The target location may be offset from the portable device. In this way, the user may be provided convenient access to a driver's door opening means.

[0016] The vehicle preferably comprises sensors for capturing data of external environmental conditions, so as to implement collision avoidance during summoning.

[0017] Two principle use scenarios/implementations are envisaged by the present disclosure. Firstly, where a car is parked in a tight parking space making it difficult to enter the car conveniently or manoeuvre out of the parking space, the system enables the user to gain access to the vehicle. Secondly, where the vehicle effectively has an automated parking valet feature or a trained parking feature. In this latter case, the disclosure accounts for when the driver/owner may not be standing precisely in a designated/rendezvous (e.g. pre-learned) spot or the driver/owner's

position cannot be precisely determined, e.g. using a global navigation satellite system (GNSS). The present system discloses a means for the vehicle to safely travel to a user over a short distance, with greater accuracy than a GNSS solution.

[0018] The secured short-range summon feature of a vehicle ensures that the vehicle can unlock and position itself precisely where a driver can reach a driver door by hand. Such functionality may be embodied by one or more of the following: [0019] i) Authentication of an owner's smart phone (e.g. as part of a PaaK system) or equivalent smart key/mobile device with compatible communication protocol carried by a user; [0020] ii) Determination of the owner's position, using the portable device as a proxy for that position, relative to a vehicle coordinate system, e.g. via ultra-wide band transceivers where position is found by triangulation/trilateration of distances between the phone and a set of nodes installed in a vehicle. In the case of trilateration (multilateration), each transceiver gives a distance to the phone. The location of the phone is estimated as that which fits best the ranges. At least three transceivers are needed to establish estimation. In the case of triangulation, a transceiver is capable of measuring the distance and the direction (azimuth) of the phone, where this measurement is called Angle of Arrival (AoA). Two distances+one AoA measurement is sufficient to estimate the location of the portable device. In another form, a combination of AoA, distance, inertial measurement unit (IMU) indication and/or limited field of view would allow the use of one transceiver to obtain an external portable device (phone) location, e.g. in a front half-circle of the vehicle; [0021] iii) Functions may be subject to proposed distance (e.g. only works up to 8-10 meters) and/or supervisory (e.g. supervised by the owner during operation) limitations; [0022] iv) Parking automation option allowing closed loop control at low speeds, and with full steering capability, where the vehicle may follow a moving user based on the current/updated/determined location of the mobile device.

[0023] The present disclosure considers integration of the above capabilities into an active vehicle relocation function using a rendezvous spot and position, e.g. determined by an algorithm using UWB positioning of an authenticated phone and environment perception based on safety sensors (e.g. selected from but not limited to ultrasound sensors, radars, vision, birds-eye-view cameras and relevant obstacle perception algorithms) to prevent collision with other vehicles or obstacles.

[0024] In a particular implementation, the system may enable automated driving in a straight line and stopping at an actively calculated rendezvous point. In other forms, the system may utilize a PaaK system and parking automation system; e.g. where there is no need of full instrumentation for high speed L2+ (level two plus autonomy in a vehicle), highway or urban pilot.

[0025] In a broad sense, the inventive concept recognizes using already available sensors/functional devices on a vehicle (e.g. an OEM vehicle as supplied to a customer) to perform additional tasks,

[0026] As mentioned, the disclosure proposes a vehicle fitted with several UWB sensors, detection or communication device configured to determine the location of a mobile/portable electronic device, as carried by a user, relative to the vehicle. The vehicle comprises a processor or control unit which may be configured for establishing a connection with the mobile device, authenticating same, waking the vehicle from a standby mode, determining or at least estimating a location of the mobile device, activating a running gear of the vehicle and guiding the vehicle to the position of the portable electronic device. Particularly, following detecting a location of the mobile device, a path is calculated to a rendezvous point (i.e. target or destination) determined based on phone's position. In this way, according to one example, at least a passenger or driver's door of the vehicle may become accessible to a user, where it was previously obscured.

[0027] In embodiments, the processor is configured for actively, e.g. continuously or at time intervals, determining a current location of the mobile device and updating the rendezvous point and path thereto.

[0028] According to a third aspect, there is provided a method of summoning a vehicle. The method at least comprises: establishing wireless communication between a communication device

onboard a vehicle and a portable electronic device external to the vehicle; authenticating the portable electronic device as authorized to summon the vehicle; determining a target location of the portable device relative to the vehicle; and autonomously driving the vehicle toward the target location in response to a summon command issued by the portable electronic device. In a particular form, the target location is updated over time, to reflect the current location of the portable electronic device.

[0029] Location is determined by trilateration or equivalent method of finding a relative location/coordinate. It is noteworthy that authorization may comprises biometric, password and/or other security features available to the portable device. In this way, a summon and unlock feature could not be activated by an unauthorized person.

[0030] According to a fourth aspect, there is provided a non-transitory computer readable medium including instructions, which when executed by one or more processors, implement the foregoing method of the third aspect. According to a further aspect, there is provided a computer software product including instructions which, when executed by one or more processors, implement the above method.

[0031] Further areas of applicability of the present disclosure will become apparent from the detailed description, the claims, and the drawings. The detailed description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0032] Illustrative embodiments will now be described with reference to the accompanying drawings in which:

[0033] FIG. 1 illustrates a vehicle incorporating a system according to the disclosure;

[0034] FIG. 2 illustrates a block diagram of one implementation of system according to the invention; and

[0035] FIG. 3 illustrates an example of a vehicle, fitted with the system, in use.

[0036] In the drawings, reference numbers may be reused to identify similar and/or identical elements.

DETAILED DESCRIPTION

[0037] The following description presents exemplary embodiments and, together with the drawings, serves to explain principles of the invention. However, the scope of the invention is not intended to be limited to the precise details of the embodiments or exact adherence with all system features and/or method steps, since variations will be apparent to a skilled person and are deemed also to be covered by the description. Terms for components used herein should be given a broad interpretation that also encompasses equivalent functions and features.

[0038] In some cases, several alternative terms (synonyms) for structural and/or system features have been provided but such terms are not intended to be exhaustive. Descriptive terms should also be given the broadest possible interpretation; e.g. the term “comprising” as used in this specification means “consisting at least in part of” such that interpreting each statement in this specification that includes the term “comprising”, features other than that or those prefaced by the term may also be present. Related terms such as “comprise” and “comprises” are to be interpreted in the same manner. Any directional terms such as “forward”, “backward”, “vertical”, “horizontal”, “up”, “down”, “sideways”, “upper” and “lower” are used for convenience of explanation usually with reference to the form shown in illustrations and are not intended to be ultimately limiting if an equivalent function can be achieved with an alternative dimension and/or direction. All directional terms are relative to each other.

[0039] The description herein refers to embodiments with particular combinations of steps or features, however, it is envisaged that further combinations and cross-combinations of compatible steps or features between embodiments will be possible. Indeed, isolated features may function independently as an invention from other features and not necessarily require implementation as a complete combination.

[0040] It will be understood that the illustrated embodiments show applications only for the purposes of explanation. In practice, the invention may be applied to many different configurations, where the embodiment is straightforward for those skilled in the art to implement.

[0041] An example vehicle configured to operate in accordance with a system/method as disclosed herein is shown by FIG. 1. The system is generally adapted for summoning a vehicle, e.g. over a short range, by a user that is external to the vehicle at the time the vehicle is summoned.

[0042] The vehicle is generally denoted **10** and, for the purposes of explanation of the present summon system, comprises a perimeter **11**, one or more communication devices or nodes **12(a-f)** and a controller **13**, e.g. an electronic control unit that is configured for utilizing the communication devices onboard the vehicle **10**. In the illustrated form, there are six communication devices/nodes including two devices proximate each of front **14** and rear **15** bumpers (e.g. **12c**, **12d** on each side of front bumper **14** and **12a**, **12f** on each side of rear bumper **15**—four in total), and one device on each side of the vehicle, e.g. proximate a door handle, namely a left side device **12b** and right side device **12e**, relative to a forward direction of travel.

[0043] External to the vehicle **10**, outside perimeter **11**, a portable communication device **16** may be located, e.g. a smart phone or equivalent device, compatible with the communication protocol of the communication devices **12**. The portable device **16** defines a proxy location for a vehicle user, e.g. a driver, when located outside of the vehicle.

[0044] In a particular embodiment, the communication devices **12** are UWB devices and the portable device **16** is enabled for communicating with such devices. In this way, the vehicle controller **13** can communicate with the portable device **16**, via one or more fixed-on-vehicle devices **12**, such as for identifying said device **16** and determining its location relative to vehicle **10**. Dotted arrows **17** represent two-way wireless communication and distance measurements so that the spatial location of device **16**, and hence an authorized vehicle operator or the like, can be determined.

[0045] Further operation of the system is best explained by reference to FIG. 2, which illustrates a block diagram of the summoning method **20**.

[0046] In a first state **21**, the vehicle may be in a standby mode, e.g. when it is parked/stationary in a parking space. At portable device **16**, a user may initiate vehicle activation, e.g. for the purposes of “summoning” the vehicle from its parking spot. A signal is sent from the portable device **16** to the controller **13**, e.g. via communication devices **12**.

[0047] In a second step **22**, the user's device **16** may be authenticated, i.e. to verify that it is associated with the owner, and therefore authorized to wake up the vehicle and activate its movement.

[0048] If summoning has been requested from device **16** (e.g. in addition to an unlocking function), at step **23** controller **13** now calculates/estimates/determines the location of the device **16**, e.g. via trilateration or an equivalent process, utilizing the vehicles sensors/devices **12**.

[0049] Once the location is known, generally within an accuracy greater than GPS capability, e.g. by virtue of the UWB communication, step **24** represents calculating a path from the stationary location, e.g. parking space, toward the portable device. The calculation may account for orientation of the device, user and/or vehicle. Preferably the target/destination at the terminus of the path will be a convenient position for a user to access a door into the vehicle, e.g. proximate device **12b** or **12e**.

[0050] The system may report to the user, via device **16**, that it is ready to activate/be summoned which could require an additional confirmation step. In any event, step **25** represents path execution

where the vehicle running gear is activated via controller **13** and movement commences, comprising forward or reverse motion dependent on the relative location and orientation of the vehicle in its parking space.

[0051] Step **26** represents termination of the summon function, where the calculated path has been executed to completion. The doors may already be unlocked or unlock upon completion depending on preference of the user. The user may now enter the vehicle, such as to take manual control or initiate other autonomous functions.

[0052] An optional function is represented by block **27**, namely a loop during path execution **25** and determining the location of the portable device at step **23**. Such a loop enables continuous or periodic recalculation of the path based upon an updated/detected location of the portable device. In other words, the system allows for deviation of the destination in the event that the user moves, such as to account for a new condition/observation. In this mode (which could be disabled/selected by the user from device **16** if desired), the vehicle may follow the user at a safe speed/minimum distance.

[0053] It is noteworthy that all movement operations of the vehicle take place at relatively low speed. Furthermore, the operational range will generally be limited to under approximately ten meters and/or within line of sight. In this way, the user may maintain oversight of the vehicle's movement and could override/stop the summoning process at any time.

[0054] It will be apparent that the summoning calculations may take advantage of steering functions of the vehicle, in order to result in the most convenient position for door access, etc.

[0055] FIG. **3** pictorially illustrates some of the concepts outlined in the steps of FIG. **2**, namely autonomous summoning of a vehicle **10** from a parking space **S**, defined by barrier lines **L**, toward a user/device **16**. In the particular example shown, a parked vehicle **P** on the right is parked over a parking space barrier line **L**, making access to a door of the vehicle tight and inconvenient. Therefore, summoning is necessary. Furthermore, the user carrying device **16** has decided to move further from parking space **S** to a second location **16a** and, thus, the path of vehicle **10** has slightly adjusted in accordance with block **27** and turned toward the user/device **16a**.

[0056] A number of extended features may be incorporated into the system disclosed herein. For example: [0057] Driver's settings (e.g. seat positions and mirror angles/preferences) may be adjusted to a user profile while the vehicle travels to the rendezvous spot feature; [0058] As the last phase of a trained parking manoeuvre or the pick up phase of an automated parking valet function: [0059] a. the vehicle may determine a rendezvous spot based on the authentication of an owner's phone (e.g. phone-as-key setup) and UWB localization (e.g. up to 8-10 m); [0060] b. the vehicle may execute parking manoeuvres to position itself according to that actively determined rendezvous spot; [0061] The vehicle may utilise the same system functionality to relocate itself at a desired relative position (rendezvous spot) to a UWB node other than (i.e. different to) the owner's personal activation device, e.g. a node installed at a vehicle charger in a garage. In this case, the desired position can therefore be determined independently of external perception sensors (e.g. radar, vision, ultrasound, bird-eye view); [0062] Driver's phone/device position may be determined by UWB system that can be additionally validated by perception sensors (e.g. radar, vision) and relevant algorithms.

[0063] The following benefits are evident in a system as disclosed herein: [0064] Rendezvous location and orientation may be actively determined based on the determined/known location of owner's mobile device, e.g. by UWB system; [0065] For maximum convenience, the rendezvous location and desired orientation can be actively controlled/adjusted by the driver by moving the phone or equivalent mobile device; [0066] Enhancement of trained parking/automated parking valet feature by an actively determined rendezvous location and orientation once driver's device is within range of UWB system;

[0067] By way of summary, the present disclosure outlines a system and method for summoning a vehicle, along with a vehicle configured for performing a summon function. Embodiments

comprise: use of a portable electronic device; a communication device or node for installation at a vehicle; a controller or processor configured for: establishing wireless communication with the portable electronic device, via the communication device or node; verifying the portable electronic device as an authorised device associated with the vehicle; determining a relative location of the portable device, via the communication device or node; calculating a path toward the relative location; and activating running gear of the vehicle to autonomously execute the path.

[0068] The inventive concept may also be summarised as a system, method or vehicle configured for utilising a portable device, e.g. mobile phone, to communicate with nodes installed at the vehicle. A relative location of the portable device is determined, via the communication nodes, which ultimately initiates autonomous driving of the vehicle toward the portable device location. A path toward the location may be updated in real time if the portable device is moved.

[0069] The term non-transitory computer-readable medium does not encompass transitory electrical or electromagnetic signals propagating through a medium (such as on a carrier wave). Non-limiting examples of a non-transitory computer-readable medium are nonvolatile memory circuits (such as a flash memory circuit, an erasable programmable read-only memory circuit, or a mask read-only memory circuit), volatile memory circuits (such as a static random access memory circuit or a dynamic random access memory circuit), magnetic storage media (such as an analog or digital magnetic tape or a hard disk drive), and optical storage media (such as a CD, a DVD, or a Blu-ray Disc).

[0070] The term “set” generally means a grouping of one or more elements. The elements of a set do not necessarily need to have any characteristics in common or otherwise belong together. The phrase “at least one of A, B, and C” should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean “at least one of A, at least one of B, and at least one of C.” The phrase “at least one of A, B, or C” should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR.

Claims

1. A system for summoning a vehicle, the system comprising: a portable electronic device; a plurality of communication devices or nodes for installation at the vehicle; and a controller configured for: establishing wireless communication with the portable electronic device via the plurality of communication devices or nodes; verifying the portable electronic device as an authorized device associated with the vehicle; determining a relative location of the portable electronic device via triangulation/trilateration of distances between the portable electronic device and the plurality of communication devices or nodes installed in the vehicle; calculating a path toward the relative location; and activating a running gear of the vehicle to autonomously execute the path.
2. The system of claim 1 wherein the controller is further configured to track any movement of the portable electronic device to determine a current relative location thereof and update the path calculation.
3. The system of claim 1 wherein the path calculation adjusts for orientation of the vehicle and/or portable device.
4. The system of claim 1 wherein the controller is further configured to relocate the vehicle to a desired location relative to a device/node other than the portable electronic device.
5. The system of claim 1 wherein the plurality of communication devices or nodes includes a plurality of ultra-wide band (UWB) devices.
6. A vehicle configured for being summoned by a user, the vehicle comprising: a plurality of communication devices or nodes including at least one ultra-wide band (UWB) device configured for wireless communication with an external portable electronic device; and a processor configured for: establishing and/or controlling wireless communication between the plurality of

communication devices or nodes and the external portable electronic device; determining a location of the external portable electronic device relative to the vehicle via triangulation/trilateration of distances between the external portable electronic device and the plurality of communication devices or nodes; calculating a path toward the relative location; and mobilizing a running gear of the vehicle to guide the vehicle along the path toward the external portable electronic device.

7. The vehicle of claim 6 wherein the processor is further configured to track any movement of the external portable electronic device, via the wireless communication, to determine a current relative location thereof and update the path calculation.

8. A method of summoning a vehicle, the method comprising: establishing wireless communication between a plurality of communication devices or nodes and a portable electronic device external to the vehicle; authenticating the portable electronic device as authorized to summon the vehicle; receiving a summon command from the portable electronic device; determining a target location of the portable electronic device relative to the vehicle by triangulation/trilateration of distances between the portable electronic device and the plurality of communication devices or nodes; and autonomously driving the vehicle toward the target location in response to the summon command issued by the portable electronic device.

9. The method of claim 8 wherein autonomous driving of the vehicle is terminated at a location where at least a door of the vehicle is accessible to a user carrying the portable electronic device.

10. The method of claim 8 wherein the target location is updated over time, to reflect a current location of the portable electronic device.

11. The method of claim 8 wherein the plurality of communication devices or nodes includes at least one ultra-wide band (UWB) device fitted to the vehicle.
